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Assessment of serum survivin in women with placenta previa and accreta spectrum: a cross-sectional study

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This study aimed to examine the relationship between maternal serum survivin levels and placental disorders by comparing these levels across cases of normal placentation, placenta previa, and placenta accreta spectrum (PAS). In this prospective cross-sectional study, we enrolled 84 pregnant women categorized into control ($n=42$), placenta previa ($n=24$), and PAS ($n=28$) groups. Serum survivin levels were quantitatively determined using ELISA, and statistical analyses were performed using ANOVA and post-hoc tests. Serum survivin levels were significantly elevated in the placenta previa (348.3 pg/mL, range: 173.0-776.4) and PAS groups (785 pg/mL, range: 324.50–1122) compared to controls (177.9 pg/mL, range: 87.3–242.0) ($p<0.05$), suggesting an association between survivin expression and abnormal placental adherence and invasion. Our findings demonstrate significant associations between elevated serum survivin levels and placental disorders, specifically placenta previa and PAS. These associations suggest a potential role for survivin in the pathogenesis of placental complications and warrant further investigation through longitudinal studies.

Keywords Placenta previa, Placenta accreta spectrum, Survivin

The placental attachment process is vital for fetal development and nourishment during pregnancy. It involves a sophisticated network of cellular and molecular mechanisms facilitating the placenta's connection to the uterine lining. Obstetric complications, such as placenta previa and the placenta accreta spectrum (PAS) disorders, pose significant challenges, frequently resulting in considerable maternal and fetal morbidity[1,2]. These conditions are primarily characterized by atypical implantation and placental invasion. Placenta previa is identified by the placenta occluding the cervical os, while PAS encompasses various conditions where the placenta inordinately adheres to the uterine wall[3].

The exploration into the molecular underpinnings of placenta previa and PAS has intensified, with the intent to refine diagnostic and therapeutic methods[4–7].

Survivin is a pivotal protein influencing cellular longevity and apoptosis[8]. Known as a cellular life sentinel, it inhibits programmed cell death. Cancer cells have exploited this characteristic to enhance their survivability, promoting tumor growth[9]. Survivin is predominantly produced by placental cytotrophoblast cells, with diminished expression in syncytiotrophoblast cells. Its primary function includes promoting cytotrophoblast proliferation and inhibiting apoptosis, which is critical for normal placental development[10]. Its role in promoting normal cytotrophoblast proliferation during pregnancy is crucial for fetal development. Nonetheless, the function of survivin in pathological pregnancies, such as hydatidiform mole and choriocarcinoma, where its levels increase, and in conditions like preeclampsia, where a decrease is noted, remains controversial[11,12]. While specific studies on circulating survivin during pregnancy are scarce, it is postulated that survivin levels might decline due to heightened apoptotic activity[13].

The objective of this study was to examine the relationship between maternal serum survivin levels and placental disorders by comparing these levels across cases of normal placentation, placenta previa, and placenta accreta spectrum (PAS). This cross-sectional study aims to explore the associations between serum survivin

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levels and placental disorders, rather than establishing its predictive capability, to better understand the role of survivin in placental pathologies.

Materials and methods

This prospective cross-sectional study was conducted at Kanuni Sultan Suleyman Research Hospital between 2021 and 2022. The cohort included pregnant women within the age range of 18 to 45 years diagnosed with placenta previa and placenta accreta spectrum disorders (PAS). A control group comprised women with physiological pregnancies. Participant recruitment was consecutive, based on admissions to the antenatal clinic or labor ward throughout the study period.

The diagnoses of placenta previa and PAS were established using ultrasound imaging. PAS was specifically suspected in cases of anechoic loss of the retroplacental area, prominent placental lacunes, and increased vascularity at the interface of the uterus and bladder. The FIGO staging criteria were applied to diagnose PAS, particularly in patients without prior cesarean section. According to the FIGO criteria, Stage 1 (placenta accreta) involves placental villi directly attached to the myometrium without intervening decidua, Stage 2 (placenta increta) involves placental villi invading the myometrium, and Stage 3 (placenta percreta) involves placental villi penetrating through the myometrium and reaching the uterine serosa or adjacent organs[14]. For patients who underwent segmental resection and hysterectomy, the resected materials were sent for pathological examination with surgical markings indicating areas suspected of placental invasion. To ensure objectivity despite these markings, two independent pathologists, who were blinded to clinical diagnoses and obstetrical outcomes, performed comprehensive examinations of the entire specimen following a standardized protocol. This protocol included systematic sampling of both marked and unmarked areas: five full-thickness sections from each placenta (including central and peripheral regions), two membrane roll sections, and three umbilical cord sections (fetal, middle, and placental ends). The pathologists independently classified PAS cases according to FIGO criteria as accreta, increta, or percreta based on the depth of placental invasion observed microscopically. Any discrepancies between the pathologists' assessments were resolved through consensus discussion.

Exclusion criteria encompassed multiple gestations, intrauterine growth restriction, and pre-existing medical conditions such as diabetes, chronic hypertension, and kidney disease.

The study protocol adhered to the Declaration of Helsinki and received approval from the Ethics Committee for Human Studies at Kanuni Sultan Suleyman Training and Research Hospital (Approval number: KAEK/2021.05.167). Written informed consent was obtained from all participants, ensuring the maintenance of privacy rights throughout the research.

Demographic data gathered included gravidity, age, parity, and medical history. Obstetric and clinical information was also compiled, including gestational age, history of previous cesarean sections, and any comorbidities.

Measurement of serum survivin

Blood samples were collected during hospital admission, specifically within the first hour of admission. The gestational ages of the participants at the time of sample collection ranged from 26 to 40 weeks. This timing was chosen to ensure consistency and to minimize any variations that might affect serum survivin levels. Serum samples were processed and stored at -80 °C until further analysis. Serum levels of survivin were quantified using a commercially available ELISA kit following the manufacturer's instructions. The assay was performed in duplicate, and the mean concentration was calculated for each sample. Serum survivin was measured with an enzyme-linked immunosorbent assay method using the commercial kit (Cat No: ab18361, Abcam Inc., UK)). The optical density (OD) was measured spectrophotometrically at a wavelength of 450 nm. The detection range was 93 pg/mL to 9600 pg/mL, and the sensitivity was 7 pg/mL.

Statistical analysis

Statistical analyses were performed using IBM SPSS (USA) and GraphPad Prism 7.0 software (GraphPad USA). Descriptive statistics were used to summarize the demographic and clinical characteristics of the study population. After the normality test, numeric variables were analyzed with ANOVA with post hoc Tukey test and Kruskal-Wallis ANOVA with post hoc Mann-Whitney test, as appropriate. To control for potential confounders, multivariate analysis was conducted using multiple regression models. This approach allowed us to adjust for variables such as age, parity, and history of cesarean section. Statistical significance was set at $p < 0.05$.

Results

Table 1 presents the baseline clinical characteristics of the women with physiological pregnancy, placenta previa, and PAS. The mean age of the PAS group was significantly higher when compared to both the control and placenta previa groups (34.0 ± 4.8 vs. 29.9 ± 5.0 and 31.9 ± 5.1, respectively; $p < 0.05$), while the mean ages of the control and placenta previa groups exhibited no significant difference ($p > 0.05$).

Among the 28 PAS cases identified through histopathological examination, 15 cases (53.6%) were classified as placenta accreta, 8 cases (28.6%) as placenta increta, and 5 cases (17.8%) as placenta percreta. Additional placental pathologies were identified during histological examination, including chronic villitis ($n = 4$, 14.3% of PAS cases), maternal vascular malperfusion ($n = 3$, 10.7%), and intervillous thrombi ($n = 2$, 7.1%). These findings were documented and considered in subsequent analyses.

Regarding parity, the placenta previa group displayed a significantly higher median parity than the control and PAS groups ($p < 0.05$). Conversely, the median parity of the control and PAS groups did not significantly differ ($p > 0.05$). The rate of multiple cesarean deliveries was notably higher in the PAS group compared to the control and placenta previa groups ($p < 0.05$).

	Controls (n = 42)	Placenta previa (n = 24)	PAS (n = 28)	Significance
Age, years	29.9±5.0[a]	31.9±5.1[a]	34±4.8[b]	0.006
Gravidity	4 (1–6)	3 (1–6)	4 (1–9)	0.058
Parity	2 (1–6)[a]	1 (0–4)[b]	2 (0–6)[a]	0.009
Number of prior cesarean delivery				0.001
0	0	14 (58.3%)	2 (7.1%)	
1	13 (31%)	9 (37.5%)	11 (39.3%)	
2	14 (33.3%)	1 (4.2%)	5 (17.9%)	
3	12 (28.5%)	0	7 (25%)	
>3	3 (7.2%)	0	3 (10.7%)	
Gestational age at delivery	38 (36–40)[a]	34 (26–40)[b]	36 (27–40)[b]	0.001
APGAR score				
at 1 min	7 (2–7)[a]	6 (2–9)[a]	6 (1–7)[b]	0.001
at 5 min	9 (6–9)[a]	9 (5–10)[a]	9 (4–9)[b]	0.001
Birth weight (g)	3085 (2190–4130)	3000 (980–4200)	2830 (1280–3370)	0.013
Obstetrical surgery				
Cesarean section	42	21	0	
Cesarean operation with uterine hemostatic suturing	0	0	9	
Cesarean with hysterectomy	0	2	11	
Cesarean with segmental uterine resection	0	1	8	
Perioperative complications				
No	42 (100%)	23 (4.2%)	22 (67.8)	
Bladder injury	0	1 (95.8%)	5 (17.9%)	
Re-laparotomy	0	0	1 (14.3%)	
Length of hospital stay	2 (1–3)[a]	2 (1–17)[b]	5 (1–37)[c]	0.001

Table 1. Baseline clinical characteristics of control, placenta previa, and PAS groups. PAS: Placenta accreta spectrum Data were expressed as mean with standard deviation, median with range, and count (%) as appropriate. When statistically significant differences exist among the study groups, they are coded with letters a, b, and c. The study groups are coded with similar letters when there is no significant difference among the study groups.

Variable	Coefficient (β)	Standard Error	t-value	*p*-value	95% Confidence Interval
Constant	-161.7580	189.881	-0.852	0.397	-539.048 to 215.532
Age	14.0472	5.920	2.373	0.020	2.285 to 25.809
Gravidity	69.0257	30.690	2.249	0.027	8.046 to 130.006
Parity	-61.1950	39.483	-1.550	0.125	-139.647 to 17.257

Table 2. Multivariable regression analysis results.

To address potential confounding factors, we conducted multivariable regression analysis to explore the relationship between maternal serum survivin levels and various baseline characteristics, including age, gravidity, parity, and previous cesarean sections. The results of the multivariable regression analysis are presented in Table 2. The analysis indicated that both maternal age and gravidity had significant effects on the serum survivin levels. Specifically, an increase in maternal age and gravidity was associated with higher serum survivin levels ($P < 0.05$). Additionally, elevated survivin levels remained significantly associated with PAS and placenta previa after adjusting for cesarean history (adjusted OR 1.82, 95% CI 1.24–2.67, $p < 0.05$). The relationship between number of previous cesarean sections and survivin levels showed a positive correlation ($r = 0.42$, $p < 0.05$), suggesting potential influence of uterine scarring on survivin expression.

The rates of hysterectomy and segmental uterine resection were also elevated in the PAS group compared to the placenta previa and control groups. Additionally, we observed a significantly lower median gestational age at delivery in both the PAS and placenta previa groups in comparison to the control group [36 (27–40) vs. 34 (26–40) and 38 (36–40), respectively; $p < 0.05$]. The median 1- and 5-minute Apgar scores of the PAS group were significantly lower than those of the other groups ($p < 0.05$). In contrast, the median 1- and 5-minute Apgar scores for the control and placenta previa groups did not exhibit significant differences ($p > 0.05$). The median birth weight of infants born to mothers in the PAS and placenta previa groups was significantly lower than that of infants born to mothers in the control group ($p < 0.05$).

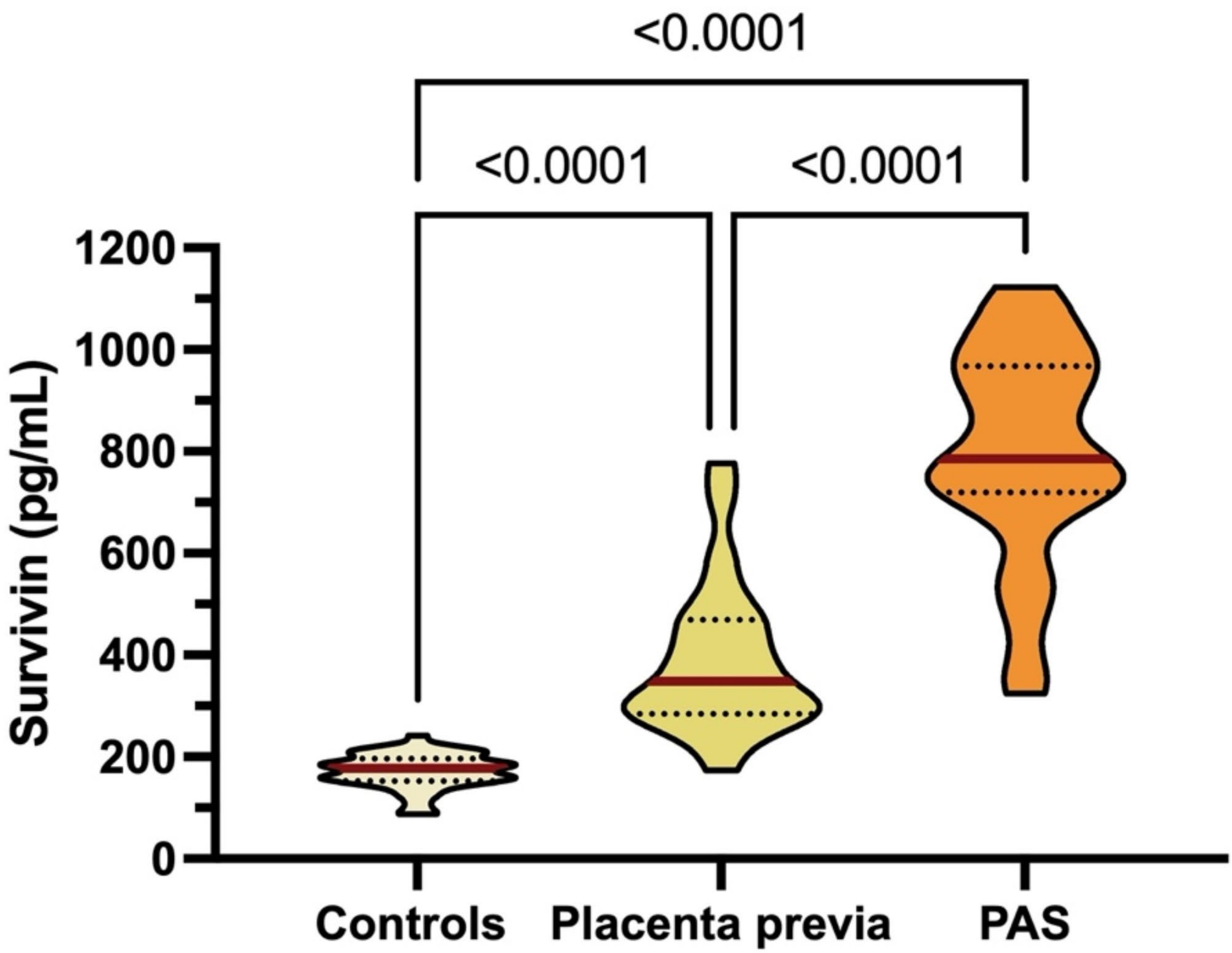

Fig. 1. Violin graph of median (25–75% interquartile range) values of serum surviving in the control, placenta previa, and PAS (placenta acreta spectrum) groups. Significances were expressed on the graph.

The median duration of hospitalization was significantly longer in the PAS group than in the placenta previa and control groups, with the placenta previa group also showing a longer hospitalization duration than the control group ($p < 0.05$).

Blood samples were collected at hospital admission (gestational age range: 26–40 weeks) to standardize the timing of survivin measurement across all groups. The median gestational age at sample collection was 34 weeks (IQR: 32–36) for PAS cases, 33 weeks (IQR: 31–35) for placenta previa cases, and 35 weeks (IQR: 33–37) for controls.

Figure 1 presents serum survivin values across all groups. Serum survivin was consistently detectable in all study groups, including normal uncomplicated pregnancies. In the control group, representing normal pregnancies, the median serum survivin level was 177.9 pg/mL (range: 87.3–242.0). These levels were significantly lower than those observed in both placenta previa (348.3 pg/mL, range: 173.0-776.4) and PAS groups (785 pg/mL, range: 324.50–1122) ($p < 0.05$), suggesting a baseline expression pattern in normal placentation. The median surviving value of the PAS group was significantly higher than those of the other groups [785 (324.50–1122) vs. 348.3 (173.0-776.4) and 177.9(87.3–242.0), respectively; $p < 0.05$]. The median surviving value of the placenta previa group was significantly higher than that of the control group ($p < 0.05$).

Discussion

The consistent detection of serum survivin in normal pregnancies in our study (median 177.9 pg/mL) provides important context for understanding its physiological role in placentation. These baseline levels likely reflect the normal activity of cytotrophoblast cells, where survivin regulates cellular proliferation and apoptosis essential for healthy placental development. The significantly higher levels observed in placental disorders (348.3 pg/mL in placenta previa and 785 pg/mL in PAS) suggest that while survivin expression is necessary for normal placentation, its overexpression may be associated with pathological placental conditions.

This study demonstrated that women with placenta previa and the placenta accreta spectrum (PAS) exhibit significantly elevated levels of maternal serum survivin compared to those with normal placentation. Survivin plays a pivotal role in modulating cell life span and preventing apoptosis, suggesting its involvement in the pathogenesis of these obstetric complications. Notably, the surge in survivin levels among PAS cases suggests a protective response to mitigate the challenges of abnormal placental attachment and invasion. This mechanism parallels survivin's function in enhancing cancer cell resilience, highlighting its critical role in pathological and physiological contexts[9].

Survivin's dual role in the placenta as an essential promoter of cellular proliferation and an inhibitor of apoptosis underlines a paradoxical contradiction: It is vital for normal placental development while contributing to placental pathologies. Our research supports the notion that survivin's upregulation is necessary for placental establishment; its overexpression may lead to conditions marked by abnormal growth and invasive behavior[11,12]. The decreased survivin levels observed in preeclampsia cases further illuminate this molecule's complex involvement in placental disorders, providing insight into diseases like placenta previa and PAS[12]. Our findings are consistent with literature that underscores the importance of survivin and related proteins in trophoblast cell proliferation and survival, highlighting the potential of targeting survivin to understand and treat placental dysfunctions[10,13,15]. For clinical practice, our study underscores the value of serum survivin as a biomarker for the early detection and management of placental diseases. Recognizing the rise in survivin levels as a harbinger of obstetric complications could lead to more effective interventions. Furthermore, this study contributes to the broader understanding of survivors' multifaceted roles, suggesting that modulating their expression could mitigate adverse outcomes in placental disorders.

We also observed a significant age difference among participants, suggesting older mothers are at higher risk of developing PAS, potentially due to age-related uterine changes. The association between multiple cesarean sections and increased PAS risk underscores the need for cautious obstetric management in such cases[16,17]. Surgical intervention rates, like hysterectomy, were also higher in the PAS group, underlining the complexity of managing PAS and the need for early detection. Significant differences were observed in gestational age at delivery and neonatal outcomes between the PAS and placenta previa groups compared to the control group. The PAS and placenta previa groups had earlier deliveries and higher rates of preterm births, along with poorer neonatal outcomes, underscoring the critical need for specialized care.

Demographically, the study highlighted the significance of maternal age and history of cesarean deliveries in the prevalence of PAS, reinforcing the need for tailored risk assessments and management plans for at-risk populations.

The multivariate regression analysis conducted in our study further strengthens the reliability of our findings. By controlling for maternal age and gravidity, we confirmed that elevated serum survivin levels observed in the PAS and placenta previa groups were not confounded by these factors. This robust analysis enhances the credibility of serum survivin as a potential biomarker for early detection and management of placental disorders.

While this study provides valuable insights into the role of maternal serum survivin levels in the pathogenesis of placenta previa and PAS, several limitations should be acknowledged. First, its cross-sectional design and limited sample size may constrain the generalizability of our findings. Future research should employ longitudinal designs and include broader demographic samples to comprehensively validate the diagnostic and therapeutic utility of survivin.

As this is a cross-sectional study, our findings demonstrate associations between serum survivin levels and placental disorders but cannot establish predictive value. The single time-point measurement of survivin levels, while providing valuable insights into the relationship between survivin and placental pathologies, limits our ability to evaluate temporal changes throughout pregnancy. Additionally, the lack of non-pregnant controls and longitudinal data prevents us from establishing predictive thresholds or determining the normal variation of survivin levels across different gestational ages.

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The potential confounding effect of previous cesarean sections and associated uterine scarring represents an important consideration in interpreting our findings. While our multivariate analysis demonstrated that the association between survivin levels and placental disorders remained significant after controlling for cesarean history, the relationship between uterine scarring and survivin expression warrants further investigation. Recent research by Sabry et al. (2022) has demonstrated survivin's role in hypertrophic scar formation, suggesting a potential mechanistic link between previous cesarean delivery and altered survivin expression.

We acknowledge that our single time-point measurement approach, while providing valuable cross-sectional data, limits our understanding of survivin's temporal dynamics throughout pregnancy. Future longitudinal studies should implement a systematic sampling protocol with measurements at key gestational periods. These should include early first trimester measurements between 11 and 13 weeks, mid-pregnancy assessments at 20–24 weeks, late pregnancy evaluation at 32–36 weeks, and measurements at delivery. This comprehensive approach would better characterize the trajectory of survivin expression and potentially identify critical windows for early detection of placental disorders.

A key limitation of our study is the absence of non-pregnant control data to establish baseline serum survivin levels. While previous research[18] has reported baseline survivin expression in non-pregnant individuals, particularly in relation to tissue healing and scarring, direct comparison of serum levels between pregnant and non-pregnant states would provide valuable context for understanding survivin's role in placental development and pathology. This limitation affects our ability to fully characterize the magnitude of survivin elevation in pregnancy-related conditions.

Additionally, we did not perform detailed analysis of survivin expression in the placenta. This analysis could offer deeper insights into the molecular mechanisms underlying placental pathologies such as placenta previa and PAS. Understanding these mechanisms could lead to targeted interventions to mitigate the risks associated with these conditions. Thus, future studies should incorporate placental survivin analysis to enhance our understanding of its role and validate the current findings.

To assess the predictive potential of survivin, future research should focus on longitudinal designs, with repeated measurements of serum survivin levels at different gestational stages. Such studies would help establish thresholds for identifying high-risk pregnancies and evaluating survivin's utility in clinical practice. Furthermore, inclusion of non-pregnant controls and examination of survivin levels in early pregnancy could provide additional insights into the role of this protein in normal and pathological placentation. Longitudinal studies could also help determine whether changes in survivin levels precede clinical manifestations of placental disorders, potentially offering new opportunities for early intervention.

Conclusions

In conclusion, our findings demonstrate significant associations between elevated serum survivin levels and placental disorders, specifically placenta previa and PAS. These associations suggest a potential role for survivin in the pathogenesis of placental complications. While our cross-sectional study provides valuable insights into the relationship between survivin and placental disorders, longitudinal studies are needed to evaluate the temporal dynamics of survivin levels and their potential utility in clinical practice. Understanding these associations contributes to our knowledge of placental pathologies and may inform future research into biomarkers for placental disorders.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Access to the data will be considered following review of a detailed research proposal to ensure appropriate use and protection of patient information.

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Author contributions

Nevin Yilmaz: Study design, data collection, analysis, and writing.Isil Turan Bakirci: Study design, analysis, and writing.Busra Sahin: Data collection, analysis.Gokhan Bolluk: Analysis, and writing.Esra Can: Analysis, and writing.Huri Dedeakayogullari: Analysis.All authors have reviewed and approved the final version of this manuscript and are accountable for the accuracy and integrity of the research.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval

This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee for Human Studies of Kanuni Sultan Suleyman Training and Research Hospital (Approval number: KAEK/2021.05.167).

Informed consent

Written informed consent was obtained from all participants, and their privacy rights were strictly maintained throughout the study.

Additional information

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